

Drugs used in gastrointestinal diseases

Learning objectives

- ✓ Describe different drugs used for treatment of gastrointestinal diseases,
- ✓ Understand the basic pharmacological principles of these drugs,
- ✓ Know the adverse effects of these drugs,

Drugs used in Acid-peptic disease

- ❖ Acid-peptic disease includes
 - ✓ peptic ulcer (gastric and duodenal),
 - ✓ gastroesophageal reflux and
 - ✓ Zollinger Ellison syndrome
- ❖ Peptic – ulcer disease is resulted from an imbalance between **cell** – **destructive effects of hydrochloric acid and pepsin** and cell-protective effects of **mucus and bicarbonate** on the other side.
- ❖ **Pepsin** is a proteolytic enzyme activated in gastric acid, also can digest the stomach wall

❖ In **gastroesophageal reflux**, acidic stomach contents enter into the esophagus causing a **burning sensation** in the region of the **heart**

✓ the common name heartburn, or other names such as indigestion, dyspepsia, pyrosis, etc

❖ **Zollinger-Ellison syndrome** is caused a tumor of gastrin secreting cells of upper part of the **small intestine** and **pancreas** characterized by excessive secretion of gastrin that stimulates gastric acid

Normal Physiology of acid secretion

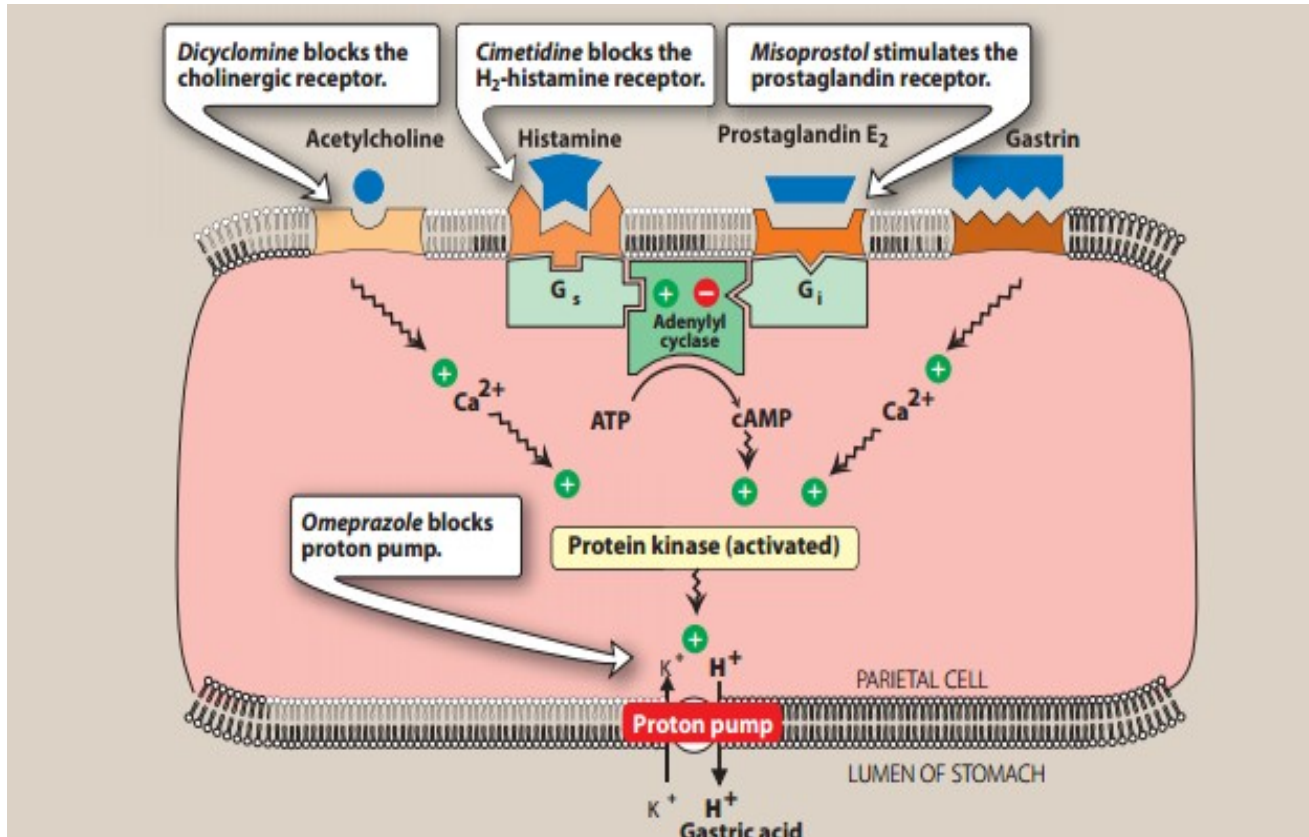


Figure: physiologic control of hydrogen ion (acid) secretion by the parietal cells of the gastric fundic glands

Physiology of HCL secretions

- ❖ The parietal cell contains receptors for
 - ✓ gastrin (CCK-B),
 - ✓ histamine (H₂),
 - ✓ and acetylcholine (muscarinic, M₃)
- ❖ When acetylcholine (from vagal postganglionic nerves) and gastrin (released from antral G cells into the blood)
- ❖ Then bind to the **parietal cell receptors**,
 - ✓ they cause an increase in cytosolic calcium, which in turn
 - ✓ stimulates protein kinases that stimulate acid secretion from a H⁺/K⁺-ATPase (the proton pump) on the canalicular surface.

- ❖ Disorders can be treated by drugs, which are able to
 - ✓ Neutralize gastric acid (HCl) e.g. magnesium hydroxide
 - ✓ Reduce gastric acid secretion e.g. cimetidine
 - ✓ Enhance mucosal defences e.g. sucralfate
 - ✓ Exert antimicrobial action against *H. pylori* e.g. clarithromycin

Gastric acid neutralizers (antacids)

- ❖ Antacids are **alkaline substances** (weak bases) that neutralize **gastric acid** (hydrochloric acid)
- ✓ They react with hydrochloric acid in the stomach
 - produce neutral or less acidic or poorly absorbed salts and raise the PH of stomach secretion
- ✓ **Drugs in this class:** sodium bicarbonate, aluminium, magnesium and calcium compounds e.g. Al(OH)_3 , MgS_2O_3 , Mg(OH)_2 , CaCO_3
- ✓ **Gastric antacids** differ in their potency, in onset of action, duration of action and adverse effects produced

Gastric acid neutralizers (antacids)...

- ❖ Magnesium compounds have
 - ✓ a relatively high neutralizing capacity,
 - ✓ rapid onset of action,
 - cause **diarrhoea** and hypermagnesemia
- ❖ Aluminium compounds generally have
 - ✓ a low neutralizing capacity,
 - ✓ slow onset of action
 - ✓ long duration of action
 - may cause **constipation**
- ❖ The most commonly used antacids, are mixtures of aluminium hydroxide and magnesium hydroxide

Gastric acid neutralizers (antacids)...

❖ Calcium compounds

- ✓ have a rapid onset of action
- ✓ may cause hypersecretion of acid (acid - rebound) and milk-alkali syndrome (hence rarely used in pepticulcer disease).

- ❖ All gastric antacids act chemically although some like **magnesium trisiolate** can also act physically.

Gastric acid neutralizers (antacids)...

❖ **Drug interaction**

- ❖ All antacids affect the absorption of the drug
 - ✓ (through reducing absorption) or
 - ✓ by increasing intragastric pH so that the drug's dissolution or solubility (especially weakly basic or acidic drugs) is altered.
- ✓ should not be given within 2 hours of doses of
 - tetracyclines,
 - fluoroquinolones,
 - itraconazole, and
 - iron.

Antisecretory drugs

- ❖ Antagonists of acetylcholine, histamine and gastrin inhibit acid secretion...Antisecretory drugs

H2-receptors blocking agents

- ✓ Drugs in this class: cimetidine, ranitidine, famotidine, nizatidine. Cimetidine

Mechanism

- ✓ Block **histamine** from binding to the parietal cell H2 receptor.
- ✓ direct stimulation of the parietal cell by **gastrin** or **acetylcholine** has a diminished effect on acid secretion in the presence of H2-receptor blockade

Table: Clinical comparisons of H₂–receptor blockers

Drug	Relative Potency	Dose to Achieve > 50% Acid Inhibition for 10 Hours	Usual Dose for Acute Duodenal or Gastric Ulcer	Usual Dose for Gastroesophageal Reflux Disease	Usual Dose for Prevention of Stress-Related Bleeding
Cimetidine	1	400–800 mg	800 mg HS or 400 mg bid	800 mg bid	50 mg/h continuous infusion
Ranitidine	4–10	150 mg	300 mg HS or 150 mg bid	150 mg bid	6.25 mg/h continuous infusion or 50 mg IV every 6–8 h
Nizatidine	4–10	150 mg	300 mg HS or 150 mg bid	150 mg bid	Not available
Famotidine	20–50	20 mg	40 mg HS or 20 mg bid	20 mg bid	20 mg IV every 12 h

Clinical uses of H₂ receptor agonists

- ❖ H₂-receptor antagonists are indicated for the following condition
 - ✓ Gastroesophageal reflux disease
 - ✓ Peptic ulcer disease—
 - ✓ Nonulcer dyspepsia
 - ✓ Prevention of bleeding from stress-related gastritis

Adverse effects of H2 receptor antagonists

- ✓ Mental status changes (confusion, hallucinations, agitation)
- ✓ Cimetidine inhibits binding of dihydrotestosterone to androgen receptors, inhibits metabolism of estradiol,
- ✓ increases serum prolactin levels
- ✓ When used long-term or in high doses,
 - it may cause gynecomastia or impotence in men and
 - galactorrhea in women.
- ❖ These effects are specific to cimetidine and do not occur with the other H2 antagonists.

Drug Interactions

- ❖ Cimetidine interferes with several important hepatic cytochrome P450 drug metabolism pathways,
 - CYP1A2,
 - CYP2C9,
 - CYP2D6, and
 - CYP3A4
- ❖ Hence, the half-lives of drugs metabolized by these pathways may be prolonged.
- ❖ Ranitidine binds **4–10 times** less than cimetidine to cytochrome P450.
 - Negligible CYP interaction occurs with nizatidine and famotidine.

- ❖ **H2 antagonists** compete with creatinine and certain drugs (eg, **procainamide**) for renal tubular secretion.
- **H2 antagonists** inhibit gastric first-pass metabolism of ethanol, especially in women.
 - Exception = famotidine

Proton-pump inhibitors (PPIs)

- ❖ Drugs in this class: omeprazole, esomeprazole, lansoprazole, dexlansoprazole, rabeprazole, and pantoprazole.
- ❖ To protect the acid-labile prodrug from rapid destruction within the gastric lumen,
 - ✓ PPIs are administered as inactive prodrugs
 - ✓ formulated as delayed release as
 - ✓ acid-resistant,
 - ✓ enteric-coated capsules or tablets.

PPIs...

Mechanism: After conversion to the active drug,

- ✓ a covalent disulfide bond with the $H^+/K^+-ATPase$, irreversibly inactivating the enzyme.
- ❖ The bioavailability PPIs decreased $\approx 50\%$ by food;
 - ✓ The drugs should be administered on an empty stomach.
- ❖ In a fasting state, only 10% of proton pumps are actively secreting acid and susceptible to inhibition.
 - ✓ PPIs should be administered approximately 1 hour before a meal (usually breakfast),
 - ✓ so that the peak **serum concentration** coincides with the maximal activity of proton-pump secretion.

Pharmacokinetics of proton pump inhibitors

Drug	pK _a	Bioavailability (%)	t _{1/2} (h)	T _{max} (h)	Usual Dosage for Peptic Ulcer or GERD
Omeprazole	4	40–65	0.5–1.0	1–3	20–40 mg qd
Esomeprazole	4	>80	1.5	1.6	20–40 mg qd
Lansoprazole	4	>80	1.0–2.0	1.7	30 mg qd
Dexlansoprazole	4	NA	1.0–2.0	5.0	30–60 mg qd
Pantoprazole	3.9	77	1.0–1.9	2.5–4.0	40 mg qd
Rabeprazole	5	52	1.0–2.0	3.1	20 mg qd

Pharmacokinetics of proton pump inhibitors...

- ❖ The drugs have a short serum half-life of about 1.5 hours,
- ❖ but acid inhibition lasts up to 24 hours owing to the irreversible inactivation of the proton pump.
 - ✓ At least 18 hours are required for synthesis of new H^+/K^+ -ATPase pump molecules

Clinical uses of PPIs

They are used in

- ❖ Gastroesophageal reflux disease
- ❖ Peptic ulcer disease
 - ✓ H.pylori
 - ✓ NSAID-associated ulcers
 - ✓ Prevention of rebleeding from peptic ulcers
- ❖ Nonulcer dyspepsia
- ❖ Prevention of stress-related mucosal bleeding
- ❖ Gastrinoma and other hypersecretory conditions

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Adverse Effects of PPIs

- ❖ Diarrhea, headache, and abdominal pain 1-5%]
- ❖ prolonged therapy decreases
 - ✓ releasing vitamin B₁₂, calcium, magnesium
- ❖ risk of hip, spine, and wrist fractures.
 - ✓ Patients who require long-term PPIs and with risk factors for osteoporosis—
 - should have monitoring of bone density and should be provided calcium supplements.
- Respiratory and enteric infections

Drug interaction

Drug Interactions

- ❖ Decreased gastric acidity **decreases absorption** of
✓ ketoconazole, itraconazole, digoxin, and atazanavir.
- ❖ omeprazole, esomeprazole, lansoprazole, and dexlansoprazole may inhibit the metabolism of **clopidogrel**, warfarin, diazepam, and phenytoin.
- ❖ Esomeprazole also may **decrease** metabolism of **diazepam**.
- ❖ Lansoprazole may enhance clearance of **theophylline**.
- ❖ **Rabeprazole** and **pantoprazole** have no significant drug interactions

Mucosal protective agents

- ❖ Their is defense mechanisms to protect itself against the noxious effects of acid and pepsin
 - ✓ Epithelial bicarbonate
 - ✓ Blood flow
 - ✓ Mucosal prostaglandins
- ❖ Agents that potentiate these mucosal defense mechanisms
 - ✓ Sucralfate
 - ✓ **Misoprostol,**
 - ✓ Bismuth compounds

Laxatives and cathartics (purgatives)

❖ Laxatives and cathartics

- ✓ are drugs used orally to evacuate the bowels or to promote bowel elimination (defecation).

❖ laxative

- ✓ implies mild effects, and eliminative of soft formed stool.

❖ cathartic

strong effects and elimination of liquid or semi liquid stool.

Example: castor oil laxative effect= 4ml

Cathartic effect = 15-60ml

Laxatives and cathartics (purgatives)...

❖ **Based on their mode of action**

- ✓ Bulk forming laxatives (E.g psyllium, bran, methylcellulose)
- ✓ largely unabsorbed from the intestine.
When
- ✓ water is added, the substances swell and become gel-like which increases the bulk of the fecal mass that stimulates peristalsis and **defecation**

Osmotic laxatives (e.g magnesium sulfate, magnesium hydroxide, sodium phosphate)

Stimulant (irritant) laxatives (cathartics) (E.g castor oil, bisacodyl, phenolphthalein, glycerine)

- Are irritant or contain an irritant substance to produce purgation
- strongest and most abused laxative products

Indications/uses of laxatives and cathartics

- ❖ To relieve constipation – bulk – forming
- ❖ To prevent straining – stool softeners
- ❖ To empty the bowel in preparation
 - ✓ bowel surgery or diagnostic procedures (saline or stimulant)
- ❖ To accelerate elimination of potentially toxic substances
- ❖ To accelerate excretion of parasite after anthelmintic drugs

Antidiarrhoeals

Diarrhoea....it is the frequent expulsion of liquid or semi liquid stools → hinders absorption of fluids and electrolytes

- ❖ Antidiarrheal drugs may be given to relieve
- ✓ the symptom (non-specific therapy) or may be
- ✓ given to treat the underlying cause of the symptom (specific therapy)

Symptomatic treatment

Opiates and opiate derivatives

- ✓ decrease diarrhea by slowing propulsive movements in small and large intestine.

E.g. diphenoxylate and loperamide

Adsorbent

- ✓ demulcent products such as kaolin – pectin preparation
- ✓ Side effect: adsorbs nutrient

Anticholinergic agents (E.g. atropine)

- ✓ decrease **abdominal cramping** and **pain** associated with diarrhea

Specific therapy

- ❖ use of antibacterial, which are recommended for use in carefully selected cases of bacterial enteritis

- ❖ Causative agent: salmonella, shigella, campylobacter

Glucose – electrolyte solution should be given in severe cases for electrolyte and fluid replacement.

- ❖ It contains :

Glucose	20 gm
NaCl	3.5gm
NaHCO ₃	2.5gm
KCl	1.5gm
Add water to	1000ml

Antiemetics

- ❖ **Antiemetics.....**Are drugs used to prevent or treat nausea and vomiting
- ✓ **Nausea....**is an unpleasant sensation of abdominal discomfort accompanied by a desire to vomit
- ✓ **Vomitnig...**is the expulsion of stomach contents through the mouth
 - Sometimes vomiting is protective mechanism which can result from various **noxious stimuli**.
- ❖ Nausea may occur without vomiting and vomiting may occur without prior nausea

Antiemetics...

Therapeutic classifications

Phenothiazines (neuroleptics) e.g. chlorpromazine

- ✓ Acts on CTZ and vomiting center
- ✓ Block dopamine receptors
- ✓ They are effective in prevention or treating nausea and vomiting induced by
 - drugs,
 - radiation therapy,
 - surgery and most other stimuli e.g pregnancy
- ✓ But ineffective in motion sickness

Antiemetics...

Antihistamines (promethazine, dimenhydrinate)

✓ Used in prevention and treatment of motion sickness

Miscellaneous antiemetics e.g Metoclopramide